



SCAMSHIELD

Detect. Explain. Protect.

AI-Powered Real-Time Protection Against Mobile Scam Messages

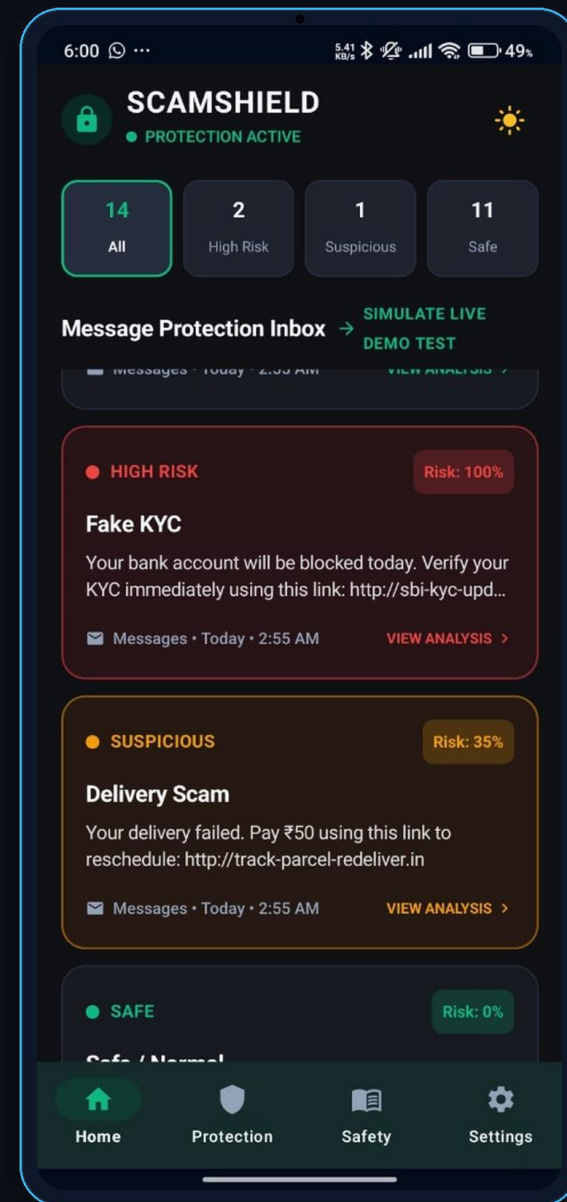
A zero-touch cybersecurity companion that automatically intercepts incoming SMS & WhatsApp threats on physical Android devices — delivering instant explainable risk analysis before victims can be deceived.

 **Zero-Copy UX** — Automatic background interception on device

 **Dual-Engine AI** — Edge heuristics + Groq LLaMA 3.3 70B cloud

 **Elderly-First** — High contrast, trilingual (EN/KN/HI) & audio TTS

Project Team: Aaron Kuriyan (Lead Developer) • @zer0neo (Collaborator)



One Message Can Cost Someone Everything.

Digital scams exploit urgency, authority, and fear — striking directly on the apps users trust most.

Fake KYC Deactivation Threat (SMS • Banking Impersonation)

"Your bank account will be blocked today. Verify your KYC immediately using this link: <http://sbi-kyc-update.net>"

UPI Collect Request Scam (WhatsApp • Financial Fraud)

"Approve collect request of ₹5,000 on PhonePe to receive your government cashback reward now."

Fake Courier & Delivery Fee (SMS • Phishing Link)

"Your parcel delivery failed. Pay ₹50 reschedule fee using this link: <http://track-parcel-redeliver.in>"

Electricity Power Cut Extortion (SMS • Utility Impersonation)

"Dear Consumer, your electricity power will be disconnected tonight at 9:30 PM due to unpaid bill. Call officer now."

Why The Crisis Is Escalating

- **Exploitation of Psychology:** Scammers bypass firewalls by manipulating emotions. Artificial urgency forces instant panic.
- **Elderly Citizens Hit Hardest:** Over 80% of elderly mobile users cannot spot spoofed URLs or deceptive domain extensions.
- **WhatsApp & SMS Dominance:** Threats arrive inside everyday personal channels alongside genuine messages from family.
- **Massive Financial Damage:** Over [INSERT VERIFIED STATISTIC] in personal savings wiped out annually across mobile banking victims.

KEY TAKEAWAY: Scam messages look identical to official notices. Traditional security expects victims to be cybersecurity experts.

Today's Protection is Too Reactive.

Existing security applications fail the very users who need protection the most.

The Traditional 6-Step Copy-Paste Burden

- 1. Message Arrives** → User receives deceptive SMS/WhatsApp threat
- 2. User Must Suspect** → Requires prior technical awareness and doubt
- 3. Manually Copy Text** → Long-press message, select text, find copy icon
- 4. Switch Application** → Exit WhatsApp, find security app, open it
- 5. Paste & Submit** → Locate text input, paste message, tap scan
- 6. Decipher Jargon** → Read technical score without actionable guidance

✗ Manual & High-Friction

Requires 6+ deliberate steps across multiple screens. Elderly and panic-stricken users simply never complete it.

✗ Protection Starts After Suspicion

If the victim trusts the impersonated bank or government notice, traditional tools are never even opened.

✗ Cryptic Threat Jargon

Displaying 'Phishing Confidence: 87.4%' confuses users instead of giving clear, plain-language directions.

✗ Invasive Privacy Trade-offs

Traditional caller-ID and spam tools harvest full address books and upload personal conversation logs to remote clouds.

THE FATAL PARADOX: Traditional tools require users to recognize the scam BEFORE they can be protected. What if protection happened automatically the instant a message arrives?

Meet SCAMSHIELD.

Turning message detection into proactive, zero-touch mobile protection.

1. AUTOMATIC DETECTION — Zero Copy-Paste Required

Native Android background service captures incoming SMS and WhatsApp messages directly via NotificationListenerService and Telephony receiver. No manual scanning, screenshots, or pasting needed.

2. INTELLIGENT DUAL-ENGINE AI — Edge Filtering + Cloud Reasoning

On-device weighted heuristics filter normal chats instantly (<5ms). Suspicious signals undergo deep Groq LLaMA 3.3 70B contextual analysis to identify deceptive intent, urgency, and fraud tactics.

3. ELDERLY-FIRST ACCESSIBILITY — Explainable, Trilingual & Voice-Enabled

High-contrast UI designed for non-technical users. Delivers plain-language guidance ('Do NOT click link', 'Do NOT share OTP'), full Kannada and Hindi localization, and Spoken Voice Warnings (TTS).



SCAMSHIELD

AI-powered mobile protection against scam messages.

From Message to Protection in Under 500ms.

A continuous, real-time background workflow engineered for speed, privacy, and explainability.

01

INCOMING MESSAGE

SMS or WhatsApp message arrives on phone

02

OS DETECTION

Android NotificationListener & SmsReceiver

03

EDGE PRE-FILTER

On-device rules filter 90%+ benign messages

04

AI ANALYSIS

Groq LLaMA 3.3 70B inspects suspicious intent

05

RISK CALIBRATION

0–100 score + threat category generated

06

USER PROTECTION

Heads-Up alert + Spoken Voice Warning (TTS)

 **SAFE (0–25%)** — Normal Everyday Chats

Messages like 'Dinner ready' or 'Call you at 5' trigger zero alarms. Stored silently in local inbox. Never leaves user's device.

 **SUSPICIOUS (26–69%)** — Unverified

Links & Deals

Dubious delivery fees, unknown lottery claims, or unofficial offers. Highlighted with amber caution card and verification guidance.

 **HIGH RISK (70–100%)** — Imminent

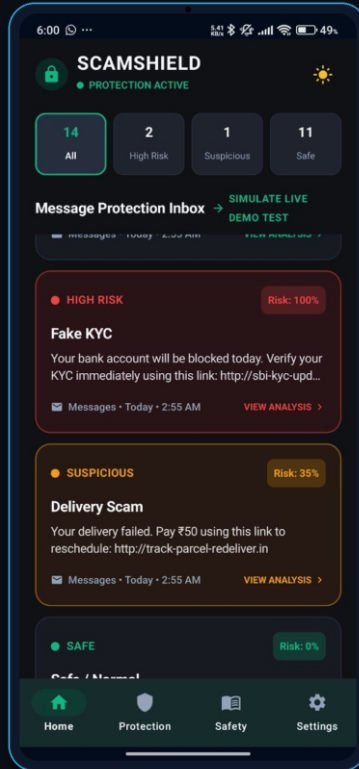
Financial Threats

Account blockage threats, OTP phishing, and UPI PIN collect traps. Triggers high-priority Heads-Up Notification and Spoken Audio Alert.

 **PRIVACY-BY-DEFAULT ARCHITECTURE:** 90%+ of harmless messages are resolved locally on the device edge. Zero plaintext personal chats are stored on the server.

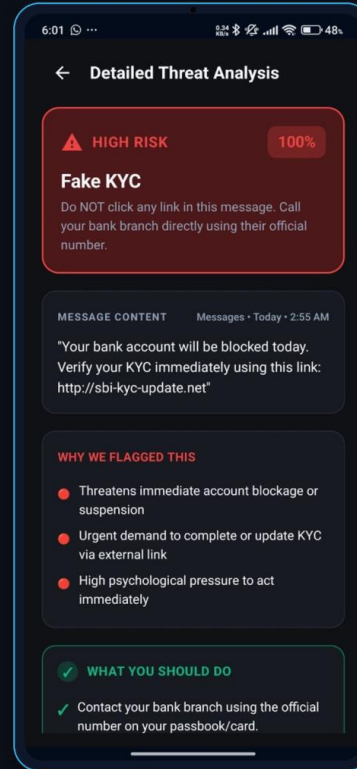
Protection That Fits Into Everyday Messaging.

Live production UI screenshots running on a physical Android device.



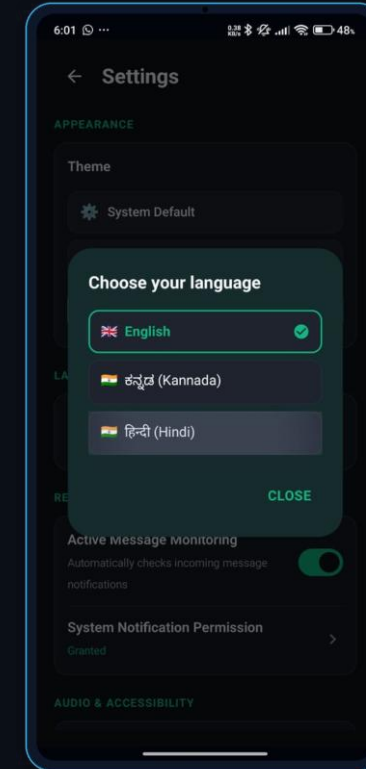
REAL-TIME DASHBOARD

Unified Message Protection Inbox with live threat counts & instant risk level chips



DETAILED THREAT ANALYSIS

100% Risk score, why it was flagged, and verified what to do / not to do steps

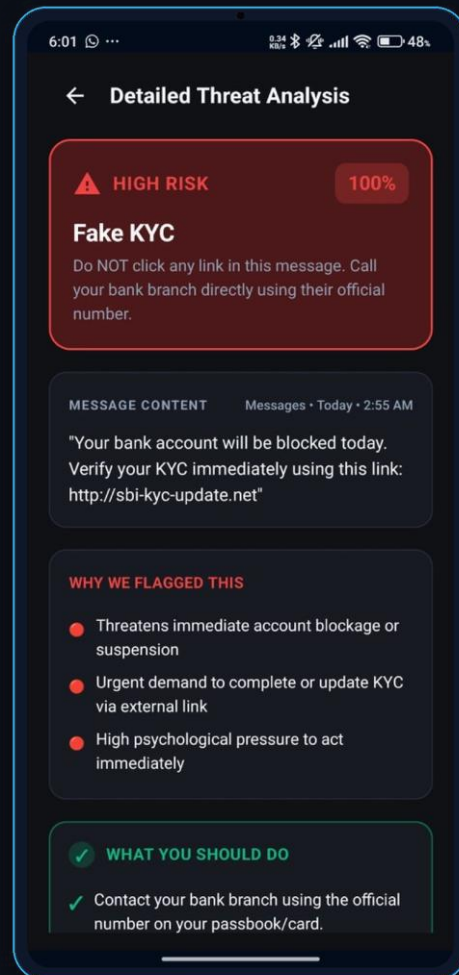


TRILINGUAL ACCESSIBILITY

English, Kannada (ಕನ್ನಡ), and Hindi (हिन्दी) system localization + active monitoring toggle

Don't Just Say 'SCAM'. Explain WHY.

Transforming threat detection into informed, confident user decisions with zero technical jargon.



🔴 HIGH RISK • 100% CONFIDENCE (Category: Fake KYC Phishing)

INTERCEPTED TEXT: "Your bank account will be blocked today. Verify your KYC immediately using this link: <http://sbi-kyc-update.net>"

▶ WHY WE FLAGGED THIS (Explainable AI Indicators)

- Threatens immediate account blockage or suspension to bypass standard caution
- Demands urgent KYC verification via an unverified external third-party domain
- Impersonates State Bank of India using deceptive URL 'sbi-kyc-update.net'
- High psychological pressure intended to force hasty action before verification

✅ ACTIONABLE CITIZEN GUIDANCE (Zero-Jargon Decisions)

WHAT TO DO: Contact your bank branch using the official customer care number on your passbook/card.

WHAT TO DO: Verify your account status strictly inside your official net banking app or branch.

WHAT NOT TO DO: Do NOT click the link. Banks NEVER send external links for mandatory KYC.

WHAT NOT TO DO: Do NOT share your OTP, UPI PIN, or passwords with anyone under any circumstance.

Why SCAMSHIELD Stands Apart.

Judging Criterion: Innovation & Originality (20 Marks) — Redefining mobile threat prevention.

FEATURE / CAPABILITY	TRADITIONAL CHECKERS	SCAMSHIELD INNOVATION
Detection Mode	Manual Copy-Paste (6+ friction steps)	100% Zero-Touch Background Interception
Protection Timing	Reactive (Only after user suspects something)	Proactive (Instant heads-up alert before click)
Analysis Output	Cryptic confidence scores ('84.2% phishing')	Plain-Language Reasons + What To Do / Not To Do
User Privacy	Cloud uploads of full messages / contacts	Zero-Surveillance Edge Gate drops 90%+ benign chats
Accessibility	English only, tiny text, complex dashboards	High-contrast, Trilingual (EN/KN/HI), Audio TTS
Offline Reliability	Crashes or fails completely without internet	Dual-Engine: Cloud Groq AI + 100% Local Fallback

UNIQUE VALUE PROPOSITION (USP): Zero-Touch Interception + Explainable Contextual AI + Zero-Surveillance Edge Gate + Elderly-First Accessibility.

Built on a Modern, Production-Grade Stack.

Inspected and verified directly from the active SCAMSHIELD codebase.

ANDROID CLIENT

Native Mobile Companion

Language: Kotlin 1.9 (Target SDK 34)

UI Framework: Jetpack Compose + Material 3

Architecture: Clean Architecture / MVVM + Coroutines & Flow

Background Service: Android NotificationListenerService & SmsReceiver

Local Database: Room SQLite Database (Encrypted metadata)

Accessibility: Android TextToSpeech (TTS) Engine

Networking: Retrofit 2 + OkHttp 3 with Dynamic Host

BACKEND & AI INFERENCE

Cloud Detection Engine

API Framework: FastAPI (Python 3.14) + Uvicorn ASGI

Cloud AI LLM: Groq Cloud SDK (Llama-3.3-70b-versatile)

Inference Speed: Sub-second contextual analysis (~350ms)

Data Contract: Pydantic v2 Type-Safe Request/Response Models

Protocol: RESTful JSON APIs with CORS Middleware

Testing: 15 Comprehensive PyTest Test Cases

Architecture: Stateless, Highly Scalable Microservice

LOCAL EDGE ENGINE

On-Device Privacy Gate

Execution: 100% On-Device Regex & Weighted Heuristics

Vectors Covered: Indian Banking, UPI PIN, KYC, Lottery, Courier

Latency: < 5ms instant execution time

False-Positive Shield: Protective Pattern Checker ('Never share OTP' -> Safe)

Zero-Surveillance: Personal chats never leave the smartphone

Deduplication: Concurrent cache prevents repeat notifications

Redundancy: Guaranteed 100% offline fallback protection

Production Architecture Under the Hood.

Judging Criterion: Technical Excellence & Implementation (25 Marks) — Robust, scalable & compliant.

SYSTEM ARCHITECTURE (EDITABLE PIPELINE)

[1] 👤 USER / INCOMING CHATS

SMS Broadcasts & WhatsApp Message Notifications



[2] 📱 MESSAGE DETECTION LAYER

ScamNotificationListenerService + Android SmsReceiver



[3] ⚡ LOCAL SCAM FILTER & DEDUP

On-Device Regex (<5ms) + Shared Hash Deduplication



[4] ☁️ BACKEND REST API

FastAPI (Python 3.14) Microservice on Uvicorn ASGI



[5] 🧠 AI SCAM ANALYSIS & REASONING

Groq Cloud SDK (LLaMA 3.3 70B) Intent & Phishing Scoring



[6] 🛡️ CLASSIFICATION & UI DISPATCH

0-100 Score + Heads-Up Alert + Trilingual UI + Voice TTS

Key Technical Challenges & Solutions

⚡ **Background OS Lifecycle:** Android frequently unbinds listeners after updates. Solved with automatic rebind requests and PackageManager state cycling.

⚡ **Cloud AI Latency & Cost:** Evaluating all chats in the cloud is slow and expensive. Dual-engine edge filter drops 90%+ benign messages on device instantly.

⚡ **Duplicate Notifications:** WhatsApp frequently updates notifications. Implemented shared cross-channel hash deduplication across SMS and WhatsApp.

⚡ **Offline Environments:** Users without active internet still need protection. Local heuristic engine provides 100% offline redundancy.

⚡ **Preventing Infinite Loops:** SCAMSHIELD warning notifications are explicitly excluded from analysis to avoid recursion.

From One Smartphone to a Safer Digital Generation.

Judging Criterion: Real-World Impact & Scalability (15 Marks) — Measurable protection & expansion roadmap.

IMMEDIATE REAL-WORLD IMPACT (LIVE TODAY)

- ✓ **Protecting Vulnerable Demographics:** Shields elderly parents and digitally inexperienced citizens from high-pressure deception.
- ✓ **Financial Loss Prevention:** Interprets fake KYC, UPI collect traps, and courier payment scams before funds are transferred.
- ✓ **Zero-Surveillance Privacy:** Maintains end-to-end user privacy by performing local edge filtering without storing personal messages.
- ✓ **Universal Voice Accessibility:** Text-to-Speech audio warnings bridge visual impairment and literacy barriers across regional demographics.

FUTURE SCOPE & SCALE (ROADMAP)

- **Broader Messaging Platforms:** Expanding listener modules to monitor Telegram, Signal, and Instagram Direct Messages.
- **Expanded Regional Languages:** Adding voice synthesis models for Bengali, Tamil, Telugu, and Marathi speakers.
- **Opt-in Family Guardian Network:** Alerting verified family guardians when a high-risk scam is detected on an elderly parent's device.
- **On-Device Small Language Models (SLMs):** Deploying quantized local models via Google LiteRT/MediaPipe for 100% offline AI inference.
- **Real-Time Phishing Domain Sandboxing:** Deep inspection of shortened links with live domain reputation and phishing site classification.

1. SINGLE USER
Real-Time Mobile Defense



2. FAMILY CIRCLE
Elderly Guardian Alerts



3. COMMUNITY
Regional Language Reach



4. NATIONWIDE
Scalable Proactive Defense

Scams Shouldn't Be the User's Problem to Solve.

Building a digital ecosystem where suspicious messages trigger immediate protection — not panic.

1. AUTOMATIC

Zero user friction. Background detection intercepts threats before victims can be deceived.

2. EXPLAINABLE

No cryptic scores. Plain-language indicators explain WHY a message is dangerous.

3. ACCESSIBLE

Built for elderly citizens first. Multilingual, high-contrast, and spoken voice guidance.

"SCAMSHIELD DETECTS.
SCAMSHIELD EXPLAINS.
SCAMSHIELD PROTECTS."



SCAMSHIELD • Hacktopia 2K26 National Hackathon Finals

Team: Aaron Kuriyan (Lead Developer) • @zer0neo (Collaborator)

Repository: Aaronkuriyan/scamshield-mobile • Ready for Live Demo

